Mohand Ouidir AMIRAT

3rd year PhD Candidate at Université Claude Bernard Lyon 1
Born on July 02, 1998 in Annaba, Algeria / Nationality: Algerian
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Research Interests

My research interests lie within the field of Control Theory for Partial Differential Equations (PDEs). I am particularly interested in the analysis and stabilization of nonlinear hyperbolic equations, with a specific focus on weakly hyperbolic systems of conservation laws. My work also involves numerical analysis and scientific computing, essentially for models arising from process engineering and fluid mechanics.

Education

PhD Candidate in Automatic Control (Specialization: Control Theory) *2023 - Pres.*
Université Claude Bernard Lyon 1 (LAGEPP) - Lyon, France

Thesis Title:	Boundary stabilization of 1-D weakly hyperbolic systems of conservation laws: application to sedimentation in wastewater treatment
Supervisors:	Vincent Andrieu, Mathieu Bajodek and Claire Valentin
Funding:	Doctoral fellowship from the French Ministry of Research (MESRI) obtained via the EEA Doctoral School (Lyon)

M.Sc. in Mathematics - Modeling and Numerical Analysis (Full 2-year program) *2021 - 2023*
Université de Montpellier - Montpellier, France

1st year M.Sc. in Applied Mathematics / Rank: 2$^{\text{nd}}$ in class *2020 - 2021*
Université Badji Mokhtar - Annaba, Algeria

B.Sc. in Mathematics / Rank: 3$^{\text{nd}}$ in class *2017 - 2020*
Université Badji Mokhtar - Annaba, Algeria

High School Diploma / Distinction: Honors (Assez Bien) *2016 - 2017*
Specialization: Technical-Mathematics - Algeria

Publications

Articles in International Journals

- **M.O. Amirat**, M. Bajodek, V. Andrieu, J. Auriol, C. Valentin. *Insights into the stability analysis of 2-by-2 linear weakly hyperbolic systems.* Accepted for publication in **Automatica**, 2026.
- J. Auriol, A. Braun, **M. O. Amirat**, M. Bajodek. Underactuated boundary control of a linearized Saint-Venant equation. *IMA Journal of Mathematical Control and Information*, 2025.
- *In preparation* – **M.O. Amirat**, V. Andrieu, M. Bajodek, C. Valentin. *Stabilization of a shock steady-state for weakly hyperbolic systems of balance laws: application to wastewater treatment.*

Conference Proceedings

- **M.O. Amirat**, V. Andrieu, J. Auriol, M. Bajodek, C. Valentin. *Boundary feedback control of a 2×2 weakly hyperbolic system: Lyapunov-based approach. IFAC-PapersOnLine, Vol. 59, Issue 8,* 2025
- **M.O. Amirat**, V. Andrieu, M. Bajodek, J. Auriol, C. Valentin. *Stability analysis of a class of 2×2 triangular hyperbolic systems. IFAC-PapersOnLine, Vol. 58, Issue 21,* 2024.

Talks & Communications

International Conferences

- 5th IFAC/IEEE-CSS Workshop on Control of Systems Governed by PDEs (CPDE 2025), Beijing, China *June 2025*
- 4th IFAC Conference on Modelling, Identification and Control of Nonlinear Systems (MICNON 2024), Lyon, France *Sept. 2024*

National Conferences

- Young Researchers in Applied Mathematics Congress (CJC-MA 2026), ENPC, France *March 2026*
- 2^{nd} Annual Congress of SAGIP, Villeurbanne, France *May 2024*

Seminars & Workshops

- Spring School on Theory and Applications of Port-Hamiltonian Systems (PHS 2025), Fraueninsel, Germany *March 2025*
- Internal Seminar LAGEPP, Université Claude Bernard Lyon 1, France *2024*

Teaching Experience

Total: 144 teaching hours *2023 – Present*

Numerical Approximation of PDEs (Master 2 ASI) *2023 – 2026*
Université Claude Bernard Lyon 1
Design and teaching of the course module "Distributed Parameter Systems" (numerical part).
- Finite Difference schemes for Heat and Advection-Diffusion equations.
- Finite Volume methods for hyperbolic systems (1D Saint-Venant).
- Implementation of Godunov schemes and approximate Riemann solvers in Matlab.

Numerical Methods for Engineers (Master 1 Automatic Control)
Sparse linear systems (direct/iterative methods) and numerical optimization.

Advanced Programming: C Language (Bachelor 3rd year)
Pointers, arrays, memory management, modular programming (makefiles), and low-level programming (bitwise operators, masks).

Introduction to Linear Control (Bachelor 2nd year)
Laplace transform, stability analysis, and elementary control (P, PI, PID).

Academic Service & Responsibilities

Co-organizer of the PhD Seminar (Café Contrôle) *2025 - Pres.*
LAGEPP, Université Claude Bernard Lyon 1 - Lyon, France
Mission: Speaker selection, logistical management, and session moderation to foster scientific exchange among PhD students in the laboratory.

Research Experience & Master Internships

2nd year M.Sc. Internship: Development of a Godunov-type Finite Volume scheme for the numerical approximation of a weakly hyperbolic system *March – Aug 2023*
Application to sedimentation in wastewater treatment
LAGEPP, Université Claude Bernard Lyon 1 - Lyon, France
Supervision: Frédéric Lagoutière, Claire Valentin and Christian Jallut

Abstract: Implementation of a Finite Volume solver for nonlinear hyperbolic systems. Development of splitting strategies. Comparison with experimental data and parameter estimation through optimization.

1st year M.Sc. Internship: Numerical methods for Hamilton-Jacobi equations and viscosity solutions
March – June 2022
IMAG, Université de Montpellier - Montpellier, France
Supervision: Jessica Guerand

Abstract: Theoretical and numerical study of Hamilton-Jacobi equations and viscosity solutions. Extension to the non-stationary case, including the analysis and implementation of a Finite Difference scheme with optimal error estimates.

Technical Skills

Operating Systems: macOS, Linux (Ubuntu), Windows

Languages: C/C++, MATLAB, Python, FreeFEM++, FEniCS

Software: ParaView, Gnuplot, Gmsh, LaTeX

Languages

French: Native/Bilingual (C2) | **English:** Advanced (C1)
Berber: Mother language | **Arabic:** Advanced